MarketSight

Stock Market Prediction System

Installation Guide

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1. System Requirements

1.1 Hardware Requirements

Component	Minimum	Recommended
CPU	2 cores	4+ cores
RAM	4 GB	8+ GB
Disk Space	10 GB	20+ GB
GPU	Not required for inference	NVIDIA GPU for model training

1.2 Software Requirements

Software	Version	Purpose
Docker	24.0+	Container runtime
Docker Compose	v2.20+	Multi-container orchestration
Git	2.40+	Source code management
Web Browser	Chrome/Firefox/Edge (latest)	Accessing the application

1.3 Optional (for development without Docker)

Software	Version	Purpose
Python	3.12	Backend development
Node.js	20+ (or Bun)	Frontend development
PostgreSQL	16	Database
uv	Latest	Python package manager

2. Installation (Docker - Recommended)

Step 1: Clone the Repository

```
git clone https://github.com/Okanagan-College-Cosc471-Winter-2026/the-project-mav
cd the-project-maverick
```

Step 2: Configure Environment Variables

Copy the example environment file and edit it:

```
cp .env.example .env
```

Edit `.env` with your settings. The key variables are:

Variable	Description	Example
`POSTGRES_USER`	Database username	`marketsight`
`POSTGRES_PASSWORD`	Database password	`your_password`
`POSTGRES_DB`	Database name	`marketsight`
`POSTGRES_HOST`	Database host	`postgres` (Docker) or `localhost`
`POSTGRES_PORT`	Database port	`5432`
`FMP_API_KEY`	Financial Modeling Prep API key	`your_fmp_key`
`SECRET_KEY`	Application secret key	`your_secret_key`

Step 3: Build and Start All Services

```
docker compose up --build
```

This will start the following services:

Service	Port	URL
Frontend (React)	5173	`http://localhost:5173`
Backend (FastAPI)	8000	`http://localhost:8000`
API Documentation	8000	`http://localhost:8000/docs`
PostgreSQL	5432	`localhost:5432`
Nginx	80/443	`http://localhost`
Airflow Webserver	8080	`http://localhost:8080`
Streamlit	8501	`http://localhost:8501`

Step 4: Verify Installation

1. Open `http://localhost:5173` in your browser - you should see the MarketSight welcome page
2. Open `http://localhost:8000/docs` - you should see the Swagger API documentation
3. Click "Start exploring" on the welcome page to access the dashboard

Step 5: Seed Market Data (First Run)

On the first run, you need to seed the database with stock data. This can be done via:

Option A: Airflow DAG

1. Open `http://localhost:8080` (Airflow)
2. Trigger the `seed_market_data` DAG

Option B: Manual Script

```
docker compose exec backend python -m app.modules.market.seed
```

3. Development Setup (Without Docker)

3.1 Backend Setup

```
# Install uv (Python package manager)
curl -LsSf https://astral.sh/uv/install.sh | sh

# Navigate to backend
cd backend

# Create virtual environment and install dependencies
uv sync

# Run database migrations
uv run alembic upgrade head

# Start the backend server
uv run fastapi dev app/main.py
```

3.2 Frontend Setup

```
# Navigate to frontend
cd frontend

# Install dependencies (using npm or bun)
npm install
# or
bun install

# Start the development server
npm run dev
# or
bun dev
```

3.3 Database Setup

```
# Start PostgreSQL (if not using Docker)
# Ensure PostgreSQL 16 is installed and running

# Create the database
createdb marketsight

# Run migrations
cd backend
uv run alembic upgrade head
```

4. Configuration

4.1 Environment Variables Reference

Variable	Required	Default	Description
`POSTGRES_USER`	Yes	-	Database username
`POSTGRES_PASSWORD`	Yes	-	Database password
`POSTGRES_DB`	Yes	`marketsight`	Database name
`POSTGRES_HOST`	Yes	`postgres`	Database host
`POSTGRES_PORT`	No	`5432`	Database port
`FMP_API_KEY`	Yes	-	FMP API key for market data
`SECRET_KEY`	Yes	-	Application secret key
`ENVIRONMENT`	No	`development`	`development` or `production`
`CORS_ORIGINS`	No	`*`	Allowed CORS origins

4.2 Nginx Configuration

The Nginx reverse proxy is configured in `docker/nginx/nginx.conf`. In production, update:

- SSL certificate paths
- Server name
- Upstream backend URL

4.3 Airflow Configuration

Airflow DAGs are located in `airflow/dags/`. Key DAGs:

- `seed_market_data` - Daily market data seeding
- `retrain_model` - Weekly model retraining
- `snapshot_datasets` - Dataset snapshot creation

5. Production Deployment (DRI Server)

For production deployment on the DRI server, refer to `docs_server_setup.md` in the project root. Key steps:

1. SSH into the DRI server
2. Clone the repository
3. Configure `.env` with production credentials
4. Run `docker compose up -d` for detached mode
5. Configure SSL certificates for Nginx
6. Use management scripts for service lifecycle:
    - Start services
    - Stop services
    - Restart services
    - Check status

6. Troubleshooting

Issue	Solution
`docker compose up` fails	Ensure Docker Desktop is running and ports are not in use
Database connection refused	Check `POSTGRES_HOST` matches your setup (`postgres` for Docker, `localhost` otherwise)
Frontend shows blank page	Clear browser cache, ensure backend is running at correct port
`alembic upgrade head` fails	Ensure database exists and connection string is correct
FMP API errors	Verify your `FMP_API_KEY` is valid and not rate-limited
Model loading fails	Ensure model artifacts exist in `ml/models/` with `meta.json`
Airflow DAGs not visible	Check that `airflow/dags/` is mounted correctly in Docker Compose
Port already in use	Stop conflicting services or change ports in `.env` / `docker-compose.yml`

7. Uninstallation

```
# Stop and remove all containers, networks, and volumes
docker compose down -v

# Remove the project directory
cd ..
rm -rf the-project-maverick
```